

Auxiliary 12 V supply for all gate drivers.

Powered devices:

UCC21551 isolated gate drivers

2EDN7533 low-side gate drivers

Input voltage range: 20 – 48 V (BOM dependent)

Output voltage: 12 V nominal (12.15 V with the selected feedback resistor values)

UVLO: 18.42V

Target switching frequency: 300 kHz

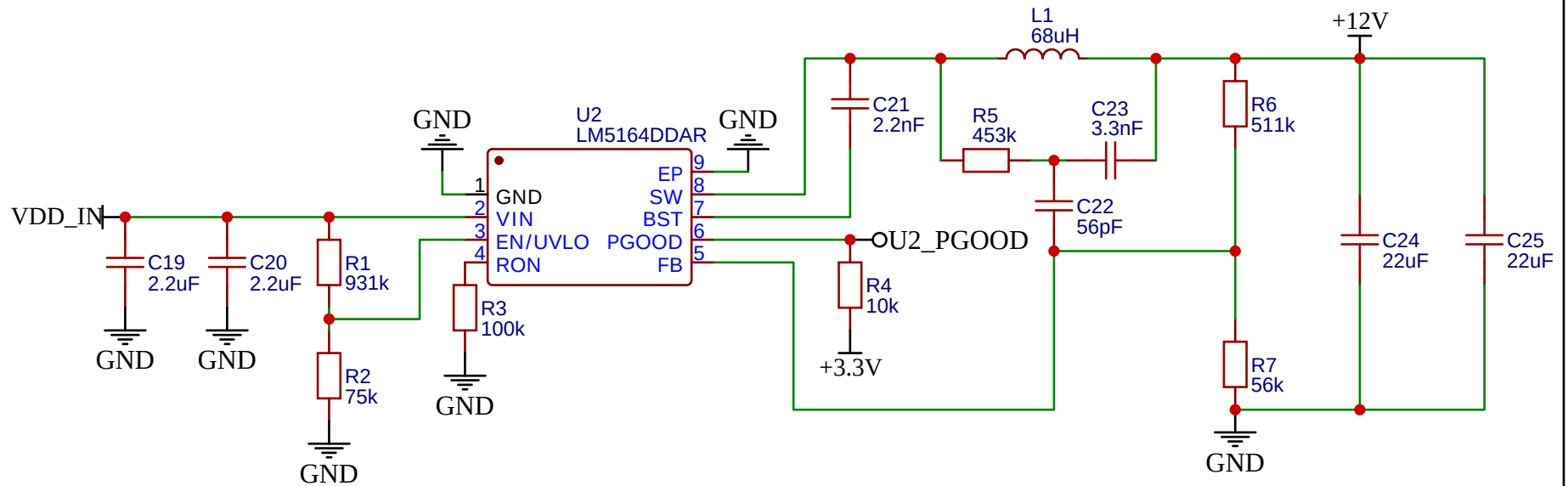
Control topology: Constant-On-Time (COT)

Power good connected to XMC4500

Input bulk capacitor with moderate ESR to damp input LC resonance and suppress VIN overshoot.

Direct VDD_IN gate drive supported up to 20 V input (via solder bridge). Above 20 V regulated 12 V supply is required.

All calculations are based on TI LM5164 datasheet (SNVSAU4D), adapted for TrinityTrack6000 HardwareControlBoard requirements.



TITLE:

Gate Driver Supply

REV: 2.0



Company:

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